

MATERNAL & CHILD NURSING HAND OUT

Fertilization and Conception

Placenta

- ▶ 8TH wk of gestation
- ▶ Cotyledons 20-25
- ▶ Respi, Renal, GI, Circulation, Endocrine
- ▶ Mechanism
- ▶ Diffusion
- ▶ Facilitated Diffusion
- ▶ Active transport
- ▶ Pinocytosis

Amniotic fluid wk gestation

- ▶ Cushion, Nutrition, Temp, Movement
- ▶ Normal: 12-15cm index
- ▶ Clear Albuminous fluid
- ▶ pH 7.0-7.25
- ▶ Rate 800-1200ml

Umbilical Cord

- ▶ AVA
- ▶ Length: 50-60cm
- ▶ Diameter: 2cm
- ▶ Wharton jelly

FETAL TESTING

Biophysical profile

- ▶ Body movement
- ▶ Muscle tone
- ▶ Breathing movement
- ▶ Amniotic fluid
- ▶ Heart beat

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Amniocentesis

- ▶ 15th & 20th week
- ▶ Color
- ▶ L/S ratio
- ▶ Bilirubin
- ▶ Fibronectin

Sandovsky & Cardiff

- ▶ Fetal movement monitor Fm/hr

Nonstress testing

- ▶ Relationship of FHR to FM
- ▶ Reactive: Rule of 15 x 15
- ▶ Non reactive: contraction stress test

Contraction stress test

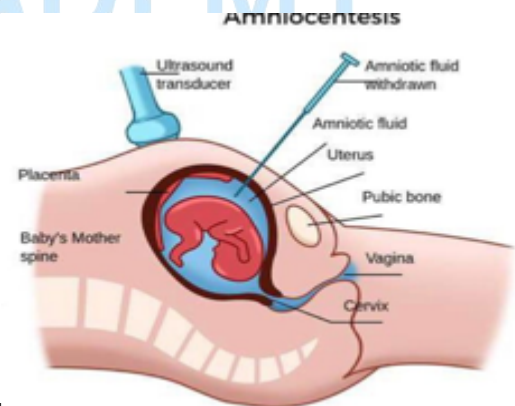
- ▶ Uterine contraction to FHR
- ▶ Normal: No late deceleration w/ contraction

Fetal Well Being test

- Amniocentesis
- L/S Ratio
- Oxytocin test
- NST
- Estriol level

Maternal serum alpha-fetoprotein

- 16wk
- 38-42mg/dl or 10-150 ng/dl



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MATERNAL CHANGES DURING PREGNANCY

Prenatal and Common Problems

1. Morning sickness
2. Urinary incontinence
3. Anemia
4. DOB
5. Backaches
6. PICA

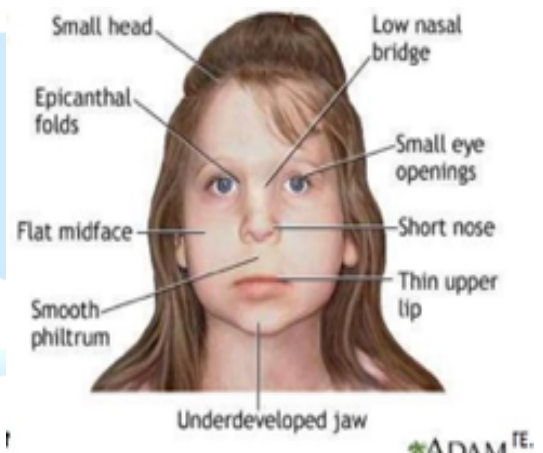


HEALTH EDUCATION

Preconception Education and Prenatal Education

Preconception Education

- Aim for a BMI of 18.5 – 24.9kg
- Obtain testing for TORCH
- Schedule dental wellness
- Ensure daily intake 400mg of Folic Acid
- Avoid alcohol and tobacco



Prenatal check up and Health Ed

Prenatal visit

- 1st 6mo – every mo
- 7-8mo – q2 weeks
- 9 mo – weekly

Weight gain

- 1st tri – 1lb/mo
- 2nd and 3rd – 1lb/weak
- Ideal 28lbs (+/- 3)

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Exercise/Travel/ Sex

- 1st tri
- 2nd tri
- 3rd tri

Nutrition

- RENI/day
- 2000 kcal/day – non pregnant
- 2300 kcal/day – pregnant + 500 if lactating

Micronutrients in a Day

- Ca - 1200mg
- Iron – 60mg
- Folate - 400mg
- Iodine - >150 lu
- Vit A – 10,000 IU (starting @ 2nd tri)

Common Teratogens

- Thalidomide
- Steroids
- Iodine
- Vit K
- Aspirin & Phenobarbital
- Streptomycin & Guinine
- Quinolones
- OHA
- NSAIDS
- Isotretinoin



Danger Signs of Pregnancy

- ANY vaginal bleeding
- Headache, dizziness, blurring of vision
- Puffiness of cheeks and face

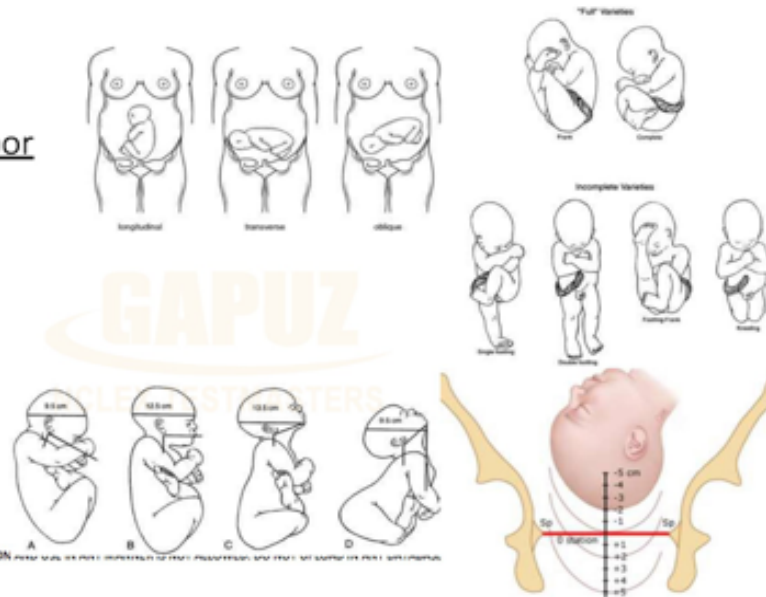
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- Paleness

LABOR AND DELIVERY

Factors Affecting Labor

- **C – Contraction**
- **L – Lie**
- **A – Attitude**
- **P – Presentation**
- **P – Position**
- **S – Station**



Stages of Labor

- 1st Stage: True labor to Cervical dilation (10cm)
- 2nd Stage: Cervical dilation to Fetal birth
- 3rd Stage: Fetal birth to Placental Delivery
- 4th Stage: 4 hours following Placental delivery

1st Stage – Dilation stage

	DILATION	FREQUENT	INTERVAL	DURATION	NXI
LATENT	3cm	10 min	5 – 30 sec	30 sec	- C - A - S - E
ACTIVE	4-7cm	5 min	3 – 5 sec	60 sec	- Conserve energy - Don't bear down - Meds - Assess FHR and Contraction
TRANSITION	8-10cm	3 min	2 – 3 sec	90 sec	- At 10cm prepare for delivery

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2nd Stage – Expulsion stage 5 Steps to Remember

1. Deliver head
2. Suction mouth then nose
3. Check for nuchal cord
4. Deliver shoulder next
5. Make sure baby has ID before leaving DR

• **Episiotomy**

- Mediolateral VS 4th degr laceration
- Medial: heals faster

• **Breathing**

- Bear down with contraction
- Inhale in the nose, exha in the mouth

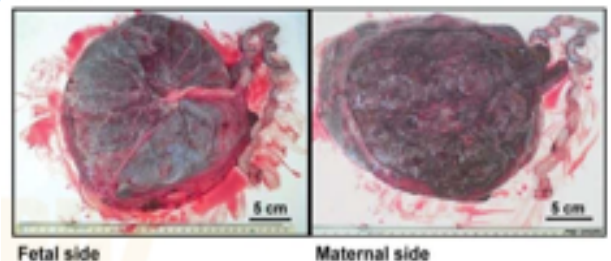


3rd Stage – Placental delivery

Placental Delivery

- Calkins
- Lengthening
- Sudden gush of blood

Term Placenta



Placental Presentation

- Shultz vs Duncan

Placental Delivery Techniques

- **Brant Andrew**
- Downward, sideways traction of cord
- **Crede**

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- Pressure on contracted fundus

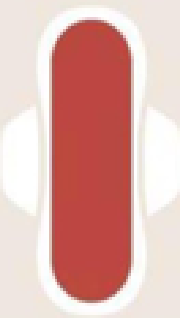
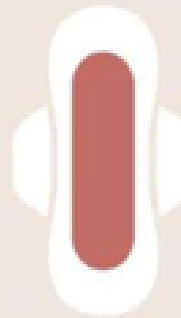
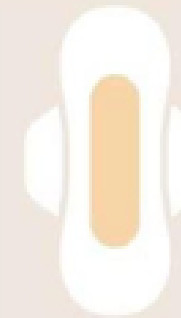
4th Stage - Recovery

- Check VS
- Check Fundus
- Check Perineal Pads
- Roll pt over and check for bleeding underneath

Post Partum Assessment

- Assess every 4 to 8 hours
- Assess for BUBBLE HEAD
- B
- U
- B
- B
- L
- E
- H
- E
- A
- D

The Stages of Lochia

LOCHIA RUBRA	LOCHIA SEROSA	LOCHIA ALBA
		
• Dark or bright red • Lasts 1-4 days • Heavy flow • Small clots are normal	• Pinkish brown • Lasts 4-12 days • Moderate flow • Less bloody, more watery	• Yellowish white or brown • Lasts 12 days - 6 weeks • Light flow or spotting • Little to no blood

Breastfeeding

- Breastfeed every 2-3 hours on average (8- 12 times/day)
- Breastfeed "on demand"
- Position the newborn "tummy to tummy"
- Ensure a proper latch
- Feed for at least 15-20 minutes per breast or until the newborn appears satisfied
- Insert a clean finger beside the newborn's gums to break suction before unlatching



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FUNDUS, LOCHIA, THROMBOPHLEBITIS

• FUNDUS/UTERUS

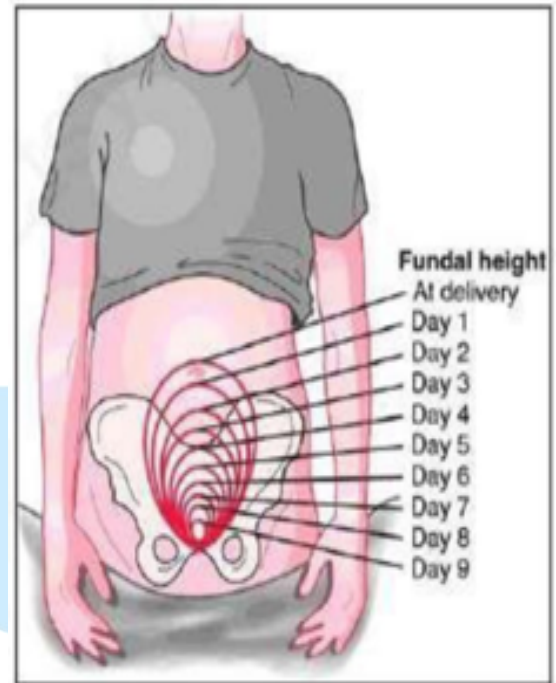
- Firm and Midline
- Involution. After delivery it is at navel

• LOCHIA

- Rubra – Red – 3days
- Serosa – Pink – 7 days
- Alba – White – 14 days

• THROMBOPHLEB

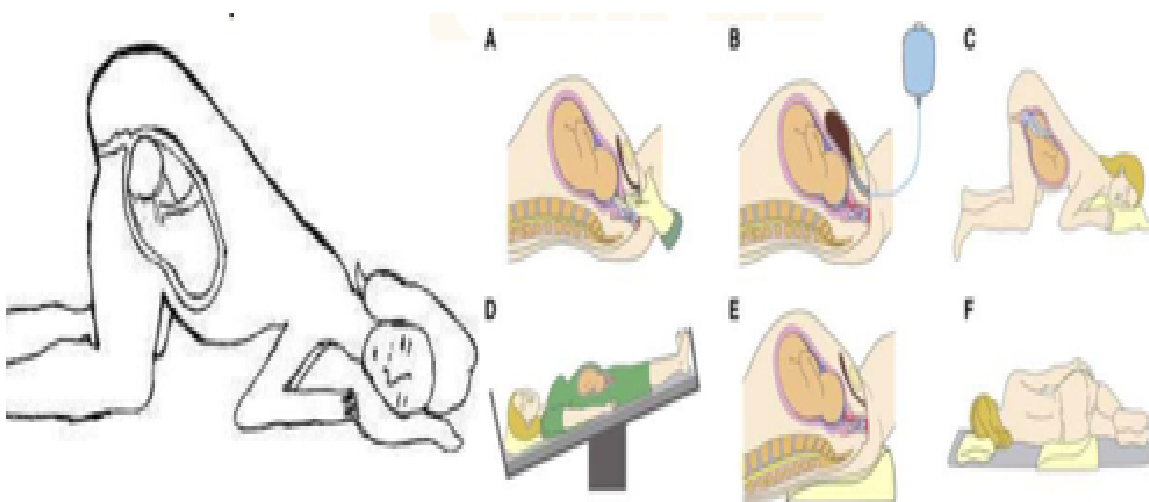
- Homan sign X
- Bilateral Calf Circumference /
- DVT – Virchows triad



Prolapse Cord

- Push the head off the cord of fetus and position mother to knee-chest or Trendelenburg

• PUSH/POSITION

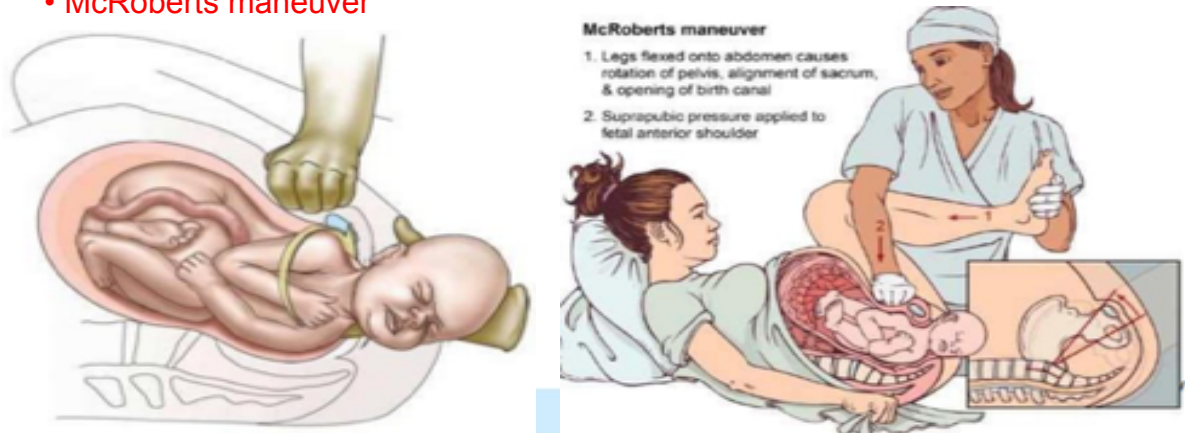


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Shoulder Dystocia

- one or both of your baby's shoulders get stuck during vaginal delivery.

- **McRoberts maneuver**



NORMAL NEWBORN

Neonatal Assessment

	0	1	2
A ppearance	Blue	Acrocyanotic	Pink
P ulse	(-)	<100	>100
G rimace	No response	Grimace	Cough/sneeze
A ctivity	Flaccid	Some flex	Full flex
R espiratory	(-)	Weak cry	Strong cry

- **Scoring**

- 0-3 – Resuscitate
- 4-6 – Airway clearance/O₂
- 7-10 – Survive

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Normal Newborn

• Fontanel

- Anterior – 18mo
- Posterior – 2-3mo

• Urine & Meconium

- Urine – 12-24hrs
- Meconium – 24-48hrs

• V/S

- Apical – 120 – 140
- RR – 30 – 60
- BP – 65/41

• Glucose

- 40 – 60mg/dl



Normal Variations

- **Milia** – white, pinhead-size, distended sebaceous gland on nose, cheek, chin and trunk
- **Epstein pearls** – Palatal cysts, white or yellow
- **Mongolian spot** – Bluish discoloration in sacral region
- **Erythema toxicum neonatorum** – Fleeting lesion – pink rash
- **Hemangioma** – Accumulation of blood vessels in skin, common birth mark



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Normal Variations

1.) HYPERBILIRUBENIMA

- Physiologic

- Appears after 24hrs

- Pathologic

- Appears w/in 24hrs

2.) Nevi (Telangiectatic nevi)

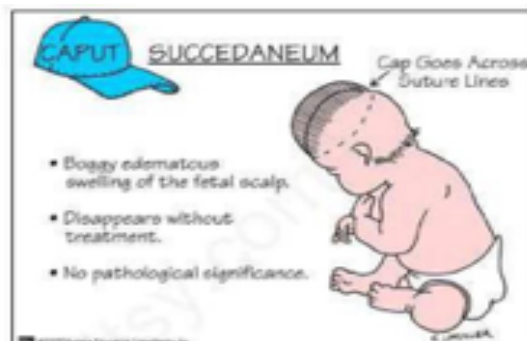
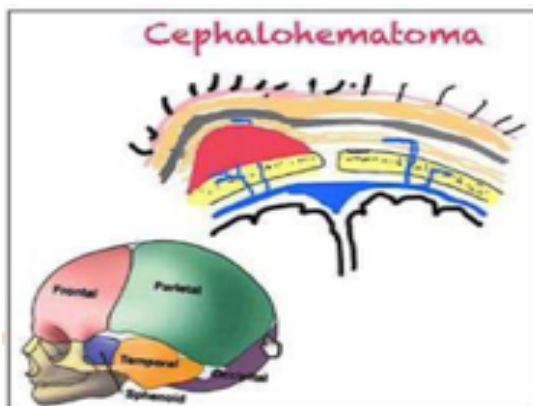
- Pink easily blanchable skin lesion. Upper eyelid, nose, upper lip, nape

3.) Port wine stain

- Non blanchable, Plexus of newly formed capillaries. Red to purple found on face and neck

CAPUT SUCCEDANEUM VS CEPHALHEMATOMA

CAPUT SUCCADANEUM	CEPHALHEMATOM
Edema/Fluid	Blood
Scalp	Periosteum / skull bone
Crosses Suture line	One or both side of the head
During birth	w/in 24 to 48hrs



Caput Succedaneum (CS)—Crosses Suture line, and Caput Symmetrical

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COMMON OB MEDS

OB Medications

- Terbutaline
- Mag Sulfate
- Pitocin
- Methergine
- Bexamethasone
- Surfactant

Tocolytics

- Stop contraction, stop labor
- Premature labor

Terbutaline (Brethine)

- S/E: Maternal Tachycardia

Mag Sulfate

- Hypermagnesemia = everything goes down
- HR, BP, Reflexes, RR, LOC

Oxytocics

- Stimulate and Strengthen labor
- Pitocin (Oxytocin)
- S/E: Uterine hyper stimulation

Methergine

- S/E: HTN

Fetal Lung Meds

• Betamethasone

- Given to mother **IM**

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- As long as baby is **in UTERO**
- S/E: increase glucose
- **Surfactant (Survanta)**
 - Given to baby via **Transtracheal** route
 - After birth

PHASE 2 : ABNORMAL OB

Complication of Labor and Delivery

1.) Hypotonic Contraction

- Cause: analgesics, bladder & bowel distention

2.) Hypertonic Contraction

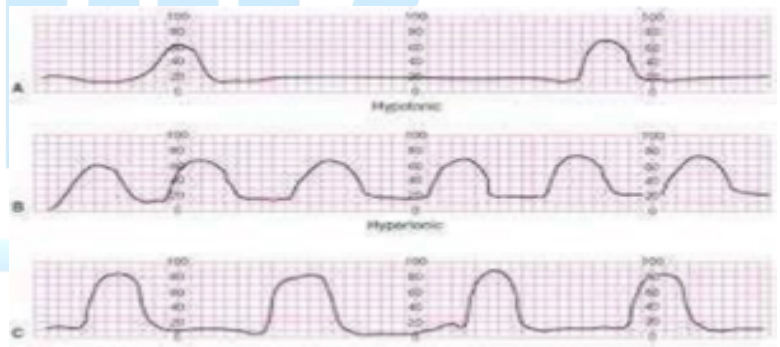
- Too much Pitocin

3.) Uncoordinated Contraction

- Ineffective child delivery

4. Precipitate labor

- Contraction is TOO strong = decrease labor time
- Injury of fetus & hemorrhage of mother



TRIMESTRAL BLEEDING

1ST Trimester

- Abortion
- Ectopic pregnancy

2nd Trimester

- Hmole
- Incompetent cervix



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3RD Trimester

- Placenta previa
- Abruptio Placenta

Type	Symptoms	Management
Threatened	CLOSED cervix	Bed rest, no coitus for 2 weks, monitor bleeding
Inevitable	OPEN cervix	Save and count pads, IV oxytocin
Incomplete	fragments retained	D&C, oxytocin IV, blood transfusion
Complete	All products expelled	oxytocin
Missed	FDIU	D&C <14 weeks, induced labor >14 weeks, monitor for infection and DIC
Habitual	Hx of 3 or more abortions, incompetent cervical os	Cervical cerclage (Shirodkar & McDonald)

Nursing Interventions

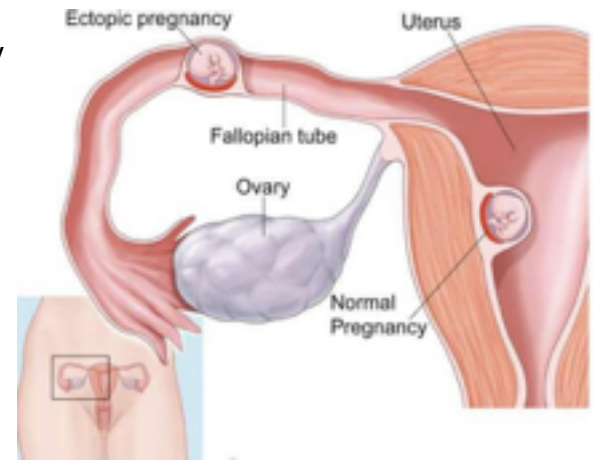
- Monitor **bleeding** & **Infection**
- LION
- Labs
- Prepare client for surgical intervention
- Administer Rhogam if mother is Rh negative



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Ectopic Pregnancy

- Fertilized ovum implanted outside uterine cavity
- s/sx – abdominal pain Vaginal bleeding (scant)
- Cullen's sign
- Shoulder pain Mgt
- culdoscopy/culdocentesis
- salpingostomy, salpingectomy
- Rhogam for Rh negative mothers
- DOC: methotrexate until (-) hCG titer (fallopian tube unruptured)



SECOND TRIMESTER BLEEDING

Hyatidiform Mole (H.Mole)

- grape-shaped vesicles in the uterus

S/Sx:

- Size not equal AOG
- No Positive signs of pregnancy
- Hypertension
- High HCG

MGT

- Shock mgt
- D & C
- No pregnancy for at least 1 year
- Methotrexate
- After extraction = Monitor Hcg titer
- Hysterectomy/chemotherapy



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THIRD TRIMESTER BLEEDING

PLACENTA PREVIA

- Low implantation of the placenta

Risk Factors

- Asian/African
- ↑parity ↑ maternal age
- Smoking/cocaine
- Prior CS/D&C

S/sx

- Painless active bleeding

MGT

- Apt/kleihauer-betke
- LION!
- Assess blood loss/shock/FHR
- No IE/Rectal exam
- Betamethasone



PRETERM LABOR

- >20 weeks and less than 37 weeks

Risk Factors

- Previous preterm labor
- Abdominal surgery
- Age
- Anomaly of fetus/placenta
- Multiple gestation

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S/sx

- Sign of true labor before term
- Vaginal spotting
- WOF! Neonatal hypoglycemia

MGT

- Tocolytic
- Betamethasone
- Survanta
- Hydration!

POST-TERM LABOR

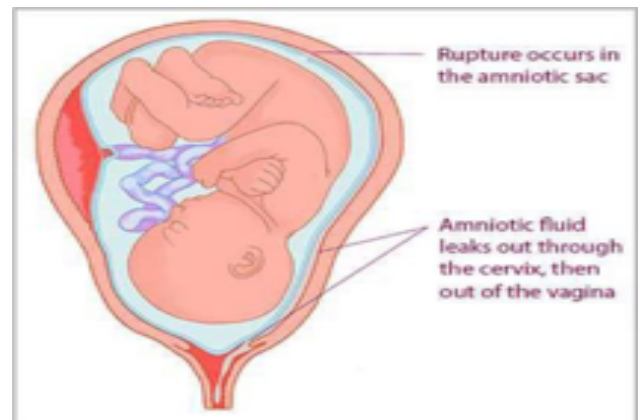
- > 38 – 42wks
- **Assessment:**
 - Dry leather skin
 - No vernix
 - Low birth wt.
 - Stiff earlobe
 - Increase crease in sole

• **Tx**

- Treat hypogly & meconium aspiration
- Development assessment

PROM

- Rupture of membrane < 37 weeks
- Complication: Cord prolapse, infection
- Assessment:
- Nitrazine paper test



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- Mgt:
- Strict bed rest
- Betametasone
- Antibx
- NO – douching, sex, vaginal exam

PIH (Pre-eclampsia and Eclampsia)

	MILD	SEVERE	ECLAMPSIA
BP	120/80 – 140/90	160/110	HPN + Convulsion
Protein	1+ - 2+ / <5g	3+ - 4+ / >5g	
Others	>2lb/wk – 2 nd tri >1lb/wk – 3 rd tri	Oliguria <500ml	

- Preeclampsia and Eclampsia Management
- L
- I
- O
- N
- Others
 - V/S
 - Bed Rest
 - Diet
 - DOC: Mag Sulfate
 - Antidote: Calcium Gluconate
 - Seizure precaution
- WOF! HELLP Syndrome